

Notes: Siani (RFS, 2025)

Raising Bond Capital in Segmented Markets

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1 Big picture

- **Question.** Why do firms often issue corporate bonds at yields above where comparable bonds trade in secondary markets, and why does this *issuance premium* become especially large in bad times?
- **Setting.** The paper studies the U.S. corporate bond primary market using bond-level issuance data, order-book information, post-issuance trading, holdings, and firm balance-sheet data. The sample spans 2002–2020 for descriptive facts and 2010–2020 for the core structural estimation sample.
- **Core challenge.** A high issuance premium could reflect many things at once: weak investor demand, adverse firm selection, information asymmetry, underwriter agency problems, or simple market-clearing price pressure when many bonds come at once.
- **Main contribution.** The paper builds and estimates a structural primary-market model with three endogenous objects: investor demand, firm bond supply, and underwriter pricing. This lets the paper quantify how much each friction contributes to issuance premiums.
- **Main baseline result.** Underwriters' investor-friendly pricing behavior is the single most important force behind the level and countercyclical behavior of issuance premiums. Investor-demand shifts and firm fundamentals matter too, but they explain less.
- **Main demand result.** Primary-market investors are quite price elastic relative to what earlier papers find in secondary markets. A 1 basis-point increase in issuance premiums raises short-term investor demand by about 5.7% and long-term investor demand by about 2.4%.
- **Main supply result.** Firms do respond to price. On average, a 10 basis-point decline in credit spreads raises issuance volume by about 1.6%, implying an average short-run supply elasticity of roughly 1.7.
- **Main underwriter result.** The estimated underwriter bargaining weight on investors is about $\hat{\eta} = 0.733$, meaning underwriters tilt price setting toward investors rather than issuers on average, and even more so in bad times.
- **Main counterfactual result.** A market-clearing primary market would generate negative average issuance premiums and cut their cyclicalities by more than half. By contrast, removing short-term investors would make the cyclicalities of issuance premiums much more severe.
- **Economic takeaway.** Primary corporate bond markets are segmented and institutionally important. The way underwriters place new bonds directly affects firms' financing costs, especially exactly when firms need funds the most.

2 Data and measurement (key objects)

2.1 Why this market matters

- U.S. firms raise more than \$1 trillion in corporate bonds each year.
- The financing cost relevant for issuers is not just the secondary-market credit spread; it is the price in the primary market where the new bond is actually sold.

- The paper emphasizes that primary and secondary markets are segmented: firms only directly transact in the primary market, while many investors do not have access to new-issue allocations.

Interpretation. The paper is not about bond valuation in general. It is about the extra wedge between the price at issuance and the price immediately after issuance, and what that wedge reveals about frictions in primary markets.

2.2 Data sources

The paper combines several data sets:

- **Mergent FISD** for bond characteristics such as size, rating, tenor, and issue date.
- **Enhanced TRACE** for post-issuance secondary-market trading and first-day spreads.
- **Credit Flow Research (CFR) and Informa Global Markets (IGM)** for order books, intra-day price talk, and issuance-stage information.
- **eMAXX** and **NAIC** for holdings and investor composition.
- **Compustat / WRDS / other standard firm data** for issuer balance-sheet and market variables.

2.3 Sample construction

There are two distinct samples.

Full sample for descriptive facts:

- U.S.-dollar nonfinancial corporate bonds,
- issue size at least \$100 million,
- July 2002 to June 2020,
- excludes exchange issues, convertibles, payment-in-kind securities, financials, utilities, and sovereign issuers.

Core estimation sample:

- 3,018 investment-grade bonds,
- September 2010 to June 2020,
- nonmissing issuance premium, underwriter, and order-book information.

2.4 Issuance premium

The core price object is the gap between the yield at issuance and the yield in the secondary market by the end of the first trading day:

$$\text{Issuance Premium} = \text{New Issue Spread} - \text{First Day Secondary Market Spread}. \quad (1)$$

The paper measures the secondary-market spread as the trade-weighted average of sell-side trades completed by the end of day 1.

Interpretation. This is the incremental annual cost of capital the firm pays because it must issue in the primary market rather than transact at the day-1 secondary-market price.

2.5 Investor demand and short-term share

- **Order book:** total quantity demanded by investors at the issuance price.
- **Short-term share:** fraction of the issue sold in the secondary market during the first week after issuance.
- **Long-term share:** one minus the short-term share.

The paper interprets this split as capturing two investor clienteles:

- **Short-term investors** who flip new bonds quickly and care strongly about capturing the issuance premium.
- **Long-term investors** who mainly buy and hold.

Interpretation. This is central to the model. The paper does not treat all primary-market investors as identical; it allows the composition of investors to shift over the cycle.

2.6 Crowdedness

The main instrument for investor demand is same-day supply pressure from other issuers:

$$Crowdedness_{uft} = \text{total bond volume issued the same day by other firms underwritten by other banks.} \quad (2)$$

It is measured at the bond-underwriter-day level and typically entered in logs (and squared logs).

Interpretation. If other firms flood the market on the same day, investors must absorb more supply. That should push up new-issue spreads, holding issuer-specific fundamentals fixed.

2.7 Firm supply variables

The firm-side estimation uses:

- the total credit spread on the new bond,
- issuer cash-to-assets,
- leverage,
- profitability,
- Tobin's Q,
- amount of debt coming due soon,
- and macro controls such as the Chicago Fed National Activity Index (CFNAI).

The key identification sample is the subset of issues for which firms change issue size during the issuance day.

2.8 Institutional facts

The paper documents several stylized facts that motivate the model.

- New bonds are commonly oversubscribed; mean order books are around 2–3 times issue size.
- Primary-market investors are a subset of all bond investors and are larger on average than secondary-market-only investors.
- Roughly 10–20% of primary-market investors flip bonds shortly after issuance.
- Short-term investor participation is higher in bad times.
- Issuance premiums spike in crises such as the GFC and COVID.

2.9 Summary statistics

From Table 1:

- Full sample mean credit spread: about 170 bps.
- Estimation sample mean issue size: about \$816 million.
- Estimation sample mean order book: about \$3.0 billion.
- Estimation sample mean issuance premium: about 5.4 bps.
- Estimation sample mean fraction sold in first week: about 21%.

Interpretation. The average issuance premium is not huge in tranquil times, but it is economically important and highly cyclical. Since aggregate issuance volume is enormous, even a few basis points matter a lot.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism

The paper’s mechanism has three layers:

- Investors are imperfectly elastic and heterogeneous.
- Firms endogenously choose issuance quantity.
- Underwriters set price by splitting surplus between firms and investors, rather than by simple competitive market clearing.

This matters because the primary market is often oversubscribed. A model that imposed $q^D = q^S$ mechanically would miss the institutional role of the underwriter.

3.2 Frictionless benchmark

For intuition, the paper starts from a frictionless benchmark in which a representative firm takes prices as given and chooses issuance quantity:

$$q^S = -\gamma r + \beta, \quad (3)$$

where r is the cost of debt and β is a reduced-form firm productivity / financing-demand shifter.

A representative investor's bond demand is:

$$q^D = \alpha r + A + v, \quad (4)$$

where A is investor wealth, v is bond-specific utility, and $\alpha > 0$ captures demand sensitivity to the spread.

Market clearing then gives

$$r_{frictionless}^* = \frac{\beta - A - v}{\alpha + \gamma}. \quad (5)$$

Interpretation. In the frictionless world, the spread is entirely pinned down by fundamentals: firm demand for funds versus investor demand for the bond.

3.3 Adding holding costs and search costs

The next step introduces two frictions:

- some investors face holding costs k , which makes them effectively short-term investors,
- investors face search costs d in finding the issue.

These frictions shift demand inward and raise equilibrium spreads. The simple market-clearing spread becomes higher because investors need compensation for these frictions.

Interpretation. Holding costs help rationalize short-term flipping, while search costs rationalize the role of underwriters as dealers who can alleviate search frictions.

3.4 Underwriter problem

The underwriter chooses the issuance spread to split gains between issuers and investors. The reduced-form problem is:

$$\max_{r, \beta_0} \pi_{UW} = \eta \pi_I(r, q(r)) + (1 - \eta) \pi_f(r, q(r)) + \beta_0 q(r), \quad (6)$$

subject to the firm's supply decision, investor participation, and enough demand to place the issue.

Here:

- $\eta \in [0, 1]$ is the weight placed on investor surplus,
- $1 - \eta$ is the weight placed on issuer surplus,
- β_0 is underwriting compensation,

- and $q(r)$ is the issue size that ends up being sold.

In unconstrained regions the solution can be written as

$$r_{UW}^* = a r_{UW, mkt}^* + b, \quad (7)$$

so the deviation from simple market clearing is a function of demand parameters, supply parameters, and the underwriter's tilt toward investors.

Interpretation. This is the core innovation of the paper. Oversubscription is not a puzzle once the underwriter is allowed to price above market clearing in order to favor investor clients.

3.5 Baseline reduced-form cyclical regression

Before the structural estimation, the paper documents that issuance premiums are countercyclical using:

$$IssPrem_{bft} = \beta EconActivity_t + X_{bft}\gamma + \varepsilon_{bft}. \quad (8)$$

$EconActivity_t$ is mainly the Chicago Fed National Activity Index (CFNAI), and controls include bond size, tenor, rating, and sometimes issuer characteristics, underwriter fixed effects, and firm fixed effects.

Interpretation. If primary markets were frictionless, weak macro conditions should already be reflected in secondary spreads. A separate rise in issuance premiums in bad times suggests extra primary-market frictions.

3.6 Investor demand equation

The core demand specification is estimated separately for short-term and long-term investors:

$$q_{bt}^h = \alpha_h^{PM} r_{bt}^{PM} + \alpha_h^{SM} r_{bt}^{SM} + \beta_h X_{bt} + \alpha_t + \varepsilon_{bt}^h, \quad h \in \{ST, LT\}. \quad (9)$$

where:

- q_{bt}^h is the quantity demanded by investor type h ,
- r^{PM} is the issuance premium,
- r^{SM} is the remaining secondary-market spread,
- X includes bond controls,
- and week fixed effects absorb low-frequency market-wide demand shifts.

Interpretation. The paper allows investors to care differently about the *issuance premium* and the rest of the credit spread. This is important because short-term investors may mostly care about the incremental spread they can capture immediately.

3.7 Demand-side first stage

To instrument for spreads in the demand equation, the paper estimates:

$$IssPrem_{uft} = \beta_1 \ln(Crowdedness_{uft}) + \beta_2 \ln(Crowdedness_{uft})^2 + \alpha_w + \alpha_f + \alpha_u + X_{uft}\gamma + \epsilon_{uft}. \quad (10)$$

The same crowdedness measure is also used to instrument for secondary-market spreads.

Interpretation. Same-day supply from other issuers shifts prices in the primary market but should not directly enter an investor’s utility for a given bond once fixed effects and controls are included.

3.8 Firm supply equation

Firm supply is built from a latent desired quantity:

$$r_b = r_b^{PM} + r_b^{SM}, \quad (11)$$

$$q^* = \gamma_r r_b + \gamma_Z Z + e, \quad (12)$$

$$q^S = \begin{cases} q^*, & \text{if } q^* > c, \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

Because issuance is lumpy and firms often do not issue at all in a quarter, the paper estimates a Tobit model with left-censoring at \$100 million.

Interpretation. Firms choose how much to issue conditional on issuing at all, and fixed issuance costs make zero issuance common.

3.9 Within-day supply identification

The key supply-side idea is that within the issuance day, firm fundamentals are effectively fixed. Some issues revise issue size within the same day as price talk updates. The paper first shows that initial price talk strongly predicts final spreads:

$$r_{b,k=4} = m r_{b,k=0} + \beta X_b + \alpha_y + \varepsilon_{bk}. \quad (14)$$

This lets the paper impute a firm’s expected final spread from initial price talk for adjusted-size issues, and then identify how quantity changes when realized spreads are better or worse than expected.

Interpretation. This is the supply-side analogue of a surprise design: the same firm, on the same day, revises quantity in response to unexpectedly favorable or unfavorable financing terms.

3.10 Underwriter second stage

The final stage estimates η with two-step efficient GMM using observed order books, spreads, and quantities together with first-stage demand and supply parameters.

Interpretation. If observed spreads are systematically above the model-implied market-clearing spread, the GMM procedure attributes that wedge to underwriter favoritism toward investors, unless the other estimated primitives can explain it.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies investor demand

Investor demand is identified from quasi-random within-week variation in same-day supply from *other* issuers using *other* underwriters.

Why this helps:

- newly issued bonds are imperfect substitutes, so extra same-day supply should force spreads up,
- week fixed effects absorb broader market-wide timing,
- firm fixed effects absorb persistent issuer characteristics,
- underwriter fixed effects absorb cross-sectional underwriter differences.

The first stage works strongly: crowded markets have higher issuance premiums and higher secondary-market spreads for new bonds.

4.2 Why reverse causality is weaker on the supply side

A naive regression of issuance on spreads would be contaminated because firms that decide to lever up might also look riskier and face higher spreads. The paper addresses this by using within-day issue-size revisions.

The identifying assumption is that over a few hours:

- firm fundamentals do not change materially,
- but investor demand conditions and price talk do move,
- so quantity revisions reflect response to financing conditions, not changing fundamentals.

Interpretation. This is much more credible than comparing different firms in different quarters because it holds fixed the issuer, the bond, and the day's underlying financing need.

4.3 Why the underwriter estimate is not just information asymmetry

The paper directly considers the classic Benveniste-Spindt interpretation that underpricing may compensate investors for information revelation. Several facts push against information asymmetry as the main explanation:

- issuance premiums are countercyclical even after controlling for a proxy for information asymmetry, namely the width of the initial spread range relative to final spread,
- the estimated underwriter tilt remains investor-friendly even for infrequent issuers, but is actually smaller there (about 0.64),

- the estimated tilt is stronger in bad times (0.77 or even 0.88 in very bad times), which lines up better with issuer desperation and underwriter bargaining power than with pure information production.

4.4 Why firm selection is not the whole story

Time-varying issuer fundamentals clearly matter, and the paper does not deny this. Instead it quantifies them. In the counterfactuals, shutting down within-firm changes in fundamentals reduces cyclical issuance-premium variation by about 17%, which means fundamentals matter but do not explain everything.

4.5 Alternative explanations the paper works through

The paper addresses several competing stories:

- **Market-wide downturns only:** week effects, macro controls, and secondary-market spreads are included.
- **Information asymmetry only:** directly proxied and tested, but not sufficient.
- **Observed bond characteristics only:** duration, rating, size, and liquidity controls are included.
- **Pure demand shocks only:** latent demand is quantified separately and explains only a modest share of time-series cyclical variation.
- **Pure firm-selection only:** firm composition and changing firm characteristics are separated in the variance decomposition.

4.6 What the paper can and cannot distinguish

What the paper does well:

- separates demand, supply, and underwriter behavior,
- quantifies the contribution of each to issuance premiums,
- and shows how investor composition stabilizes primary markets.

What it does not fully do:

- fully microfound all investor heterogeneity,
- cleanly separate all forms of information asymmetry from all other latent-demand components,
- or claim Lucas-critique-proof counterfactuals when the market structure is radically changed.

5 Tables and figures

	Count	Mean	Std Dev	Percentile 1	Median	Percentile 99
Full sample						
Credit spread (bps)	4,433	170	115	30	140	582
Coupon	4,433	3.68%	1.39%	.88%	3.62%	7.62%
Yield to maturity	4,431	3.72%	1.39%	.91%	3.67%	7.75%
Amount (\$MM)	4,433	798	666	250	600	3000
Tenor (years)	4,433	12.3	9.6	2.0	10.0	40.0
Credit rating	4,433	14.3	2.8	7.0	15.0	22.0
Crowdedness (\$Bn)	4,357	3.0	3.5	0	1.9	16.6
Order book (\$Bn)	3,706	3.0	2.2	.5	2.4	11.0
Issuance premium (bps)	4,415	6.4	8.9	-7.7	4.6	43.6
Pct sold first week	4,433	.22	.15	.02	.19	1.00
Estimation sample						
Credit spread (bps)	3,018	145	81	30	130	425
Coupon	3,018	3.43%	1.12%	.88%	3.45%	5.95%
Yield to maturity	3,018	3.47%	1.12%	.94%	3.49%	5.99%
Amount (\$MM)	3,018	816	637	250	600	3000
Tenor (years)	3,018	12.6	9.5	2.0	10.0	31.0
Credit rating	3,018	15.3	2.2	12.0	15.0	22.0
Crowdedness (\$Bn)	3,018	3.3	3.7	.0	2.1	17.5
Order book (\$Bn)	3,018	3.0	2.1	.5	2.4	11.0
Issuance premium (bps)	3,018	5.4	7.8	-7.7	4.1	38.6
Pct sold first week	3,018	.21	.15	.02	.18	1.00

This table reports summary statistics for (1) the full sample of bonds and (2) the estimation sample, which includes only bonds that have at least one investment-grade (IG) credit rating. "Credit rating" is numerically coded as per Table 1A.11 in the Internet Appendix. "Crowdedness" is computed as the total volume of bonds issued by other firms, underwritten by other broker-dealers on the same day. "Issuance premium" is computed as the difference between the new issuance credit spread and the trade-weighted average of sell-side trades completed by the end of the first day post issuance. I omit extreme values with changes of greater than 300 basis points (bps). "Pct sold first week" is the ratio of total sell orders in the secondary market in the first week following issuance (as reported by Enhanced TRACE) to the size of the bond at issuance (as reported by FISD). Source: Mergent FISD, IGM, CFR, eMaxx, and Enhanced TRACE.

5.1 Table 1: summary statistics

Read: The typical investment-grade issue in the estimation sample is about \$816 million, has a mean order book near \$3 billion, an average issuance premium of 5.4 bps, and about 21% of the issue sold in the first week.

Economic message. These summary statistics establish three core facts: primary issues are large, they are commonly oversubscribed, and a nontrivial minority of investors are short-term flippers. Those are exactly the ingredients the structural model needs.

5.2 Figure 1 and Table 2: issuance premiums rise in bad times

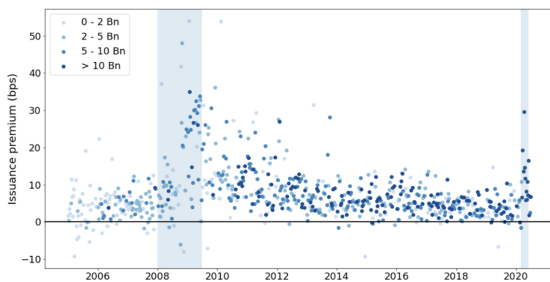


Figure 1
First-day credit spread changes
I plot the time series of weekly averages in issuance premiums for newly issued bonds. The issuance premium is defined as the credit spread difference, in basis points, between new-issue credit spread and the volume-weighted average credit spread on sell trades reported in TRACE completed by end of the first day following issuance. Shaded regions are January 2008–June 2009 (i.e., the GFC) and March–May 2020. Darker dots indicate weeks with greater issuance volumes. Source: Enhanced TRACE and Mergent FISD.

Table 2
Issuance premiums are higher in bad times

	(1) Baseline	(2) Issuer controls	(3) UW info	(4) Firm FE	(5) Est sample
Economic activity	-1.016*** (0.349)	-1.060*** (0.351)	-1.063*** (0.340)	-1.031*** (0.326)	-1.010*** (0.247)
Issuance range / spread			-7.183** (2.763)	-6.439** (2.536)	
Credit rating (log)	-14.54*** (1.137)	-16.18*** (1.371)	-15.32*** (1.422)	-13.15*** (3.316)	-22.57*** (7.273)
Bond size (log)	.718* (0.364)	.777** (0.344)	.854* (0.330)	1.274*** (0.368)	1.424*** (0.395)
Tenor (log)	-1.435*** (0.249)	-1.453*** (0.243)	-1.726*** (0.286)	-1.607*** (0.220)	-0.817*** (0.179)
Debt / assets		-2.594 (1.797)	-2.064 (1.697)	-2.904 (4.325)	-3.224 (5.525)
Cash / assets		1.244 (2.178)	1.611 (2.104)	8.720 (7.176)	2.784 (3.748)
Operating profit / assets		34.31*** (11.73)	33.48*** (11.61)	23.55 (14.13)	35.31*** (12.16)
Firm FE			✓	✓	✓
Underwriter FE				✓	✓
Observations	17,412	17,412	17,391	17,353	8,595
R-squared	.140	.144	.157	.484	.465

The dependent variable is issuance premium, measured in basis points. The independent variable of interest is economic activity, as measured by the CFNAI monthly index, which is designed to be mean zero with a standard deviation of one. Bond controls include issuer credit rating (log), size of bond (log), and tenor in years. Firm controls in Regressions (2) through (4) include the prior quarter cash to total assets ratio, total debt to total assets ratio, and operating profit to total assets ratio. Regressions (3) and (4) control for bond-level issuance range as a proportion of the final issuance credit spread. Regressions (3) and (4) include underwriter fixed effects (FE). Regression (4) includes firm fixed effects. Observations are at the bond-underwriter level. Regression (5) restricts the sample to only those observations included in the demand estimation (est). Standard errors are clustered at the month and underwriter level.

Read: Figure 1 plots weekly issuance premiums and shows large spikes during the GFC and the COVID stress episode. Table 2 formalizes this by regressing issuance premiums on economic activity.

Economic message. A deterioration in macro conditions raises issuance premiums even after controlling for bond characteristics, issuer characteristics, information-asymmetry proxies, underwriter fixed effects, and even firm fixed effects. So the premium is not just a by-product of who issues in bad times.

5.3 Table 3 and Figure 2: short-term investors matter more in bad times

Table 3
Increased short-term investor participation in bad times

	(1) Short-term share	(2) Firm fundamentals	(3) Demand-side effects
Economic activity	-0.00175*** (0.000336)	-0.00213*** (0.000370)	-0.00138*** (0.000434)
Bond size (log)	-0.0255*** (0.00241)	-0.0237*** (0.00233)	-0.0280*** (0.00220)
Tenor (log)	0.0477*** (0.00132)	0.0489*** (0.00134)	0.0491*** (0.00127)
Credit rating (log)	-0.214*** (0.0179)	-0.293*** (0.0212)	-0.308*** (0.0204)
Cash / assets		-0.0278 (0.0240)	0.0211 (0.0270)
Debt / assets		-0.141*** (0.0165)	-0.216*** (0.0203)
TEDRATE			0.0215*** (0.00628)
Dealer intermediated volume ratio			-0.0100*** (0.00132)
Firm FE	✓	✓	✓
Underwriter FE	✓	✓	✓
Observations	11,340	10,134	9,975
R-squared	.408	.408	.419

The dependent variable is the share of short-term investors for each bond, measured as the selling activity in the first week following issuance divided by the size of the bond issuance. The independent variable of interest is the CFNAI monthly index, a proxy for economic activity. Bond controls include size of bond (log), tenor in years, and issuer credit rating (log). Regression (2) adds probability of default, prior quarter cash to assets, and prior quarter leverage. Regression (3) adds the TED spread (TEDRATE) and the dealer intermediated volume ratio, computed as the ratio of weekly buy volume from customers to weekly buy volume from dealers. All of the regressions include firm and underwriter fixed effects. Observations are at the bond-underwriter level and include all bonds in the estimation sample. Standard errors are clustered at the underwriter level.

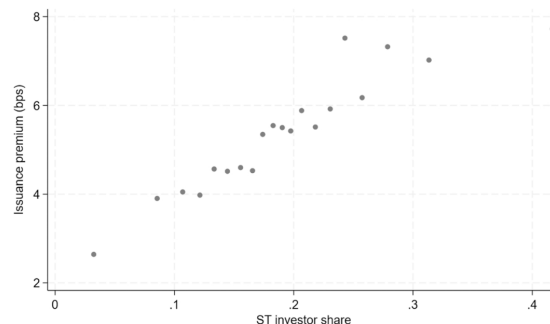


Figure 2
The relationship between higher issuance premiums and more short-term investors
The figure shows a binned scatter plot of the share of the bond sold within the first week on issuance premium, conditional on the short-term (ST) share being between 0 and 1. It includes controls for issuer credit rating, bond tenor, bond size (log), and U.S. Treasury yields as well as year, firm, and underwriter fixed effects. Analysis includes all observations of the estimation sample. Source: cMAXX, Enhanced TRACE, and Mergent FISD.

Read: Table 3 shows that weaker economic activity is associated with a larger short-term investor share. Figure 2 shows a positive cross-sectional relation between the issuance premium and the share sold in the first week.

Economic message. This pattern supports the idea that short-term investors are attracted by issuance-premium opportunities, especially in stressed markets. They behave like marginal elastic capital that steps in when spreads rise.

5.4 Table 4 and Table 5: demand identification and elasticities

Read: Table 4 shows that same-day issuance by other firms raises both issuance premiums and secondary-market spreads, validating crowdedness as a supply shifter. Table 5 then reports the IV demand estimates.

Economic message. The key numbers are large. A 1 bp rise in issuance premiums raises short-term demand by 5.7% and long-term demand by 2.43%, while the same change in the remainder of the secondary-market spread has almost no effect. Primary-market investors care especially about the incremental spread created at issuance, and short-term investors care the most.

5.5 Figure 3, Table 6, and Table 7: supply-side evidence

Read: Figure 3 shows initial price talk is a strong predictor of final spreads. Table 6 uses within-day size adjustments to estimate firm supply elasticities, and Table 7 shows how those elasticities vary across rating groups and time periods.

Table 4
Price impacts of supply shocks in primary markets

	(1) Issuance premium (bps)	(2) SM credit spread (bps)
Amount issued by other firms (log)	2.012*** (0.340)	4.751** (2.025)
Amount issued by other firms (log sq)	-0.857*** (0.148)	-3.023*** (0.874)
U.S. Treasury yield	-0.468*** (0.144)	-1.314 (0.904)
Bond size (log)	1.016*** (0.160)	15.89*** (1.095)
Credit rating (log)	-18.03*** (3.125)	-358.1*** (17.83)
Tenor (log)	-0.428*** (0.0974)	33.25*** (0.997)
Bidask	-0.415 (0.362)	1.961 (1.636)
Underwriter FE	✓	✓
Week FE	✓	✓
Issuer FE	✓	✓
Observations	11,340	11,340
R-squared	.613	.881

The dependent variable in the first regression is the issuance premium. The dependent variable in the second regression is the secondary market (SM) credit spread on the newly issued bond, measured in basis points. The independent variables of interest are the amount issued by other firms, underwritten by other banks, in the same day (both logged and logged squared). Credit spreads and crowdedness measures are winsorized at the 1% level. Controls include the U.S. Treasury yield for the duration of the bond, size of bond (log), issuer credit rating (log), tenor in years (log), and the monthly weighted average bid-ask spread (Bidask). The model includes underwriter, issuer, and week fixed effects. Observations are at the bond-underwriter level. Standard errors are clustered at the underwriter level.

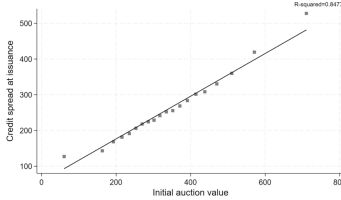


Figure 3
Correlation of initial price talk with final treasury spreads
The y-axis shows the initial announced credit spread for a given bond. The x-axis shows the credit spread for a given bond. The model includes year fixed effects and controls for issuer credit rating, bond size (log), and bond size. Source: IOM, CFR, and Morgan IFSO.

Table 6
Firm supply estimates (standard t-stat)

	(1) All issuance	(2) Amount issued by IG firms (log)	(3) Amount issued by HY firms (log)
PM credit spread (bps)	-0.00162 *** (0.000179)	-0.00154 *** (0.000166)	-0.00145 *** (0.000168)
U.S. Treasury yield	0.850*** (0.160)	1.159*** (0.342)	0.762*** (0.248)
Credit rating	-1.477*** (1.230)	-6.147*** (3.543)	-2.790 (1.750)
Cash/Assets last qtr	-4.584*** (1.091)	-10.51*** (3.468)	1.990 (2.516)
CFNAI	-0.149*** (0.0551)	-0.209*** (0.0530)	-0.152 (0.108)
Leverage last qtr	-1.650*** (0.467)	-7.176*** (2.281)	-2.866** (1.426)
ROA last qtr	10.09** (4.573)	27.13** (13.64)	-0.325 (4.521)
Amount due in 3 months	0.0225** (0.0111)	0.0429** (0.0170)	-0.0218 (0.0380)
Tobin's Q	0.000582 (0.000489)	0.000886 (0.000719)	0.000307 (0.00115)
/	14.79** (9.712)	31.19 (20.64)	14.72 (11.41)
varc.logant	17.511 (12.022)	5.499	
Observations			1

This table covers the sample bonds issued 2000-2020. Observations are at the firm-quarter level. I include firm fixed effects. Standard errors are clustered at the firm level. The standard t-stat estimation is left-censored at log of \$100 million. The first regression is estimated in a simultaneous within-bond estimation where standard errors are clustered at the bond level. The dependent variable is logged quarterly issuance volume plus one.

Table 5
Primary market estimates

	(1) Qd short-term (log)	(2) Qd long-term (log)	(3) Qd(ST) / Qd(LT) (log)
Issuance premium (bps)	0.0570*** (0.0134)	0.0243*** (0.00627)	0.0326*** (0.0119)
SM credit spread (bps)	0.00181 (0.00248)	0.000468 (0.000941)	0.00134 (0.00210)
U.S. Treasury yield	-0.0477** (0.0210)	-0.0213* (0.0107)	-0.0264 (0.0206)
Bond size (log)	0.484*** (0.0467)	0.648*** (0.0195)	-0.164*** (0.0379)
Credit rating (log)	0.565 (0.882)	0.444 (0.359)	0.121 (0.710)
Tenor (log)	0.367*** (0.0891)	-0.0229 (0.0309)	0.390*** (0.0766)
Bidask	-0.385*** (0.0328)	-0.0630*** (0.0137)	-0.322*** (0.0329)
Underwriter FE	✓	✓	✓
Week FE	✓	✓	✓
Issuer FE	✓	✓	✓
Observations	11,340	11,340	11,340
F-statistic	808.0	1462.0	667.6

This table covers sample bonds issued 2010-2020 with the share of short-term investors between 0 and 1. Observations are at the bond-underwriter level. Controls include issuance amount (log), issuer credit rating (log), tenor in years (log), and the monthly weighted average bid-ask spread. Instruments include the amount of bonds issued on the same day by other firms and underwritten by other broker-dealers (log), average rating, and the cash balances of same-day bond issuers. Credit spreads and crowdedness measures are winsorized at the 1% level. The model includes bank fixed effects to account for cross-sectional variation in underwriter balance sheets and variation in expected rationing, week fixed effects to absorb trends in demand for capital, and firm fixed effects to account for cross-sectional variation in unobserved firm characteristics. Standard errors are clustered by bank.

Table 7
Firm supply estimation (standard t-stat)

	(1) BIB-rid	(2) A-rid	(3) 2010-2019	(4) 2008-2009	(5) 2020 Pre-Fed	(6) 2020 Post-Fed
Quantity (log)	-0.00428*** (0.000408)	-0.00560*** (0.000491)	-0.00162*** (0.000222)	-0.00146*** (0.000506)	-0.00288*** (0.000613)	-0.00537*** (0.000677)
Credit spread (bps)	0.0121*** (0.00105)	0.0109*** (0.00067)	0.0215*** (0.00256)	0.0173*** (0.00283)	0.0113*** (0.00394)	0.00859*** (0.00267)
varc.logant	17.511 (12.022)	5.499				
Observations	589	225	1875	210	65	57

The table covers bonds that are in the main estimation sample. Observations are at the bond-issuer stage level. The dependent variable for all of the columns is issuance volume, in log. Standard errors are clustered at the bond level. The standard t-stat estimation is left-censored at log of \$100 million. I include within-bond fixed effects.

Economic message. Firms do respond to prices, but not infinitely. A 10 bp decline in spreads raises issuance about 1.6% on average. Investment-grade firms are more elastic than high-yield firms, and firms are less elastic in crisis-like conditions. This is exactly why underwriters can extract more from issuers in bad times.

5.6 Table 8: underwriter favoritism

Read: The baseline estimate is $\hat{\eta} = 0.733$. The estimate is around 0.70 in good times, 0.77 in bad times, and 0.88 in very bad times.

Economic message. Underwriters systematically tilt toward investors, not issuers, and that tilt becomes stronger when firms are more desperate to raise capital. This is the central structural result of the paper.

Table 8
Second-stage estimates

$\hat{\eta}$	$\hat{\eta}^{Good}$	$\hat{\eta}^{Bad}$	$\hat{\eta}^{VeryBad}$	$\hat{\eta}^{Infrequent}$	$\hat{k}$	$\hat{\lambda}_1$	$\hat{\lambda}_2$
0.7331 (0.00866)	0.6979 (0.00875)	0.7693 (0.00864)	0.8845 (0.00844)	0.6356 (0.00856)	0.0002 (0.05299)	-0.0016 (0.00059)	4.1532 (0.10197)

This table reports the second-stage points estimates computed using a two-stage efficient GMM as described in detail in B.1.2. Bootstrapped standard errors are reported in parentheses. "Good" indicates estimates on the sample when economic activity is above average; "Bad" indicates estimates on the sample when economic activity is below average; "VeryBad" indicates estimates on the sample when economic activity is below the 25th percentile; "Infrequent" indicates the firm has not issued in over 30 days.

Table 9
Variance decomposition

	(1) Firm composition	(2) Firm propensity to issue	(3) Market characteristics	(4) Latent demand
Percent of variation	0.507*** (0.0344)	0.104*** (0.0156)	0.283*** (0.0346)	0.105*** (0.0161)

This table reports the decomposition of the issuance premium variance into variations arising from (1) firm selection into issuance; (2) time-varying firm characteristics; (3) market and bond characteristics that enter into investor demand, including secondary market credit spreads; and (4) the latent component of investor demand. Robust standard errors are reported in parentheses.

Table 10
Counterfactual magnitudes of issuance premium cyclicalty

	(1) Baseline	(2) Neg shock	(3) Pos shock	(4) Avg firm fund	(5) LT inv	(6) Latent demand	(7) Mkt	(8) Mkt w srch
Economic activity	-1.047*** (0.219)	-1.152*** (0.222)	-0.934*** (0.217)	-0.864*** (0.324)	-1.716*** (0.245)	-0.955*** (0.127)	-0.393*** (0.0602)	-0.420*** (0.0597)
Firm controls	✓	✓	✓	✓	✓	✓	✓	✓
Bond controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Underwriter FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	8,690	8,690	8,690	8,690	8,690	8,690	8,690	8,690

The outcome variable is issuance premium, measured in basis points. The dependent variable is the monthly CFNAI index. The model includes firm and underwriter fixed effects (FE). The controls include prior-quarter leverage, cash-to-assets ratio, and profitability as measured by operating income over total assets. The bond controls include tenor (log), rating (log), bond size (log) and the ratio of initial price to final price during issuance. Standard errors are clustered at the underwriter level. Neg = negative; Pos = positive; Avg = Average; LT inv = long-term investors; Mkt = Market equilibrium; Mkt w srch = Market with search.

5.7 Table 9: variance decomposition

Read: The paper decomposes the cross-sectional variance of issuance premiums into firm composition, time-varying firm propensity to issue, market characteristics, and latent investor demand.

Economic message. About 51% comes from which firms are in the issuing pool, 10% from time-varying firm characteristics, 28% from market and bond characteristics, and 11% from latent demand. So both firm-side and market-side forces matter, but the cross section is dominated by who issues.

5.8 Table 10: counterfactual cyclicalty

Read: Table 10 compares the model's baseline cyclicalty to counterfactual economies. Fixing firms' nonprice fundamentals at their average values reduces cyclicalty by about 17%. Eliminating short-term investors amplifies cyclicalty by 64%. Shutting down latent demand reduces cyclicalty by about 9%. Market-clearing pricing cuts cyclicalty by more than half.

Economic message. Two forces stand out. First, shifting firm fundamentals in bad times matter because distressed firms want to issue more. Second, underwriter pricing is what turns that higher willingness to pay into especially large primary-market spreads. Short-term investors work in the opposite direction by cushioning the system.

6 Short summary

- The paper studies the wedge between primary-market and secondary-market bond yields, the issuance premium.
- It documents that issuance premiums are strongly countercyclical and spike in crises.
- It builds a structural model with heterogeneous investors, endogenous firm bond supply, and underwriter discretion in pricing.
- Primary-market investor demand is much more elastic than secondary-market demand, especially for short-term investors.
- Firms do increase issuance when spreads fall, but their short-run elasticity is moderate.
- Underwriters favor investors over issuers on average, with a bargaining weight around 0.733, and this tilt is stronger in bad times.

- Firm composition explains much of the cross-sectional variation in issuance premiums, but countercyclical time-series variation is importantly amplified by underwriter pricing.
- Short-term investors stabilize the primary market by entering more when issuance premiums are high.
- In a counterfactual market-clearing world, average issuance premiums would be negative and much less cyclical.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the corporate bond primary market is not just a passive reflection of secondary-market prices. New-issue spreads are determined in a segmented market where investor demand, issuer supply, and underwriter incentives all matter. The most important new result is that underwriter favoritism toward investors is a first-order driver of issuance-premium levels and cyclicity.

7.2 Economic intuition

Bad times do two things at once. They make investors more cautious, but they also make firms more eager to raise funds. If pricing were purely competitive, spreads would rise only to clear the market. In reality, underwriters have discretion and use it in a way that tends to preserve investor rents. Because firms are also less price elastic in bad times, that discretion bites hardest precisely when firms most need credit. At the same time, high premiums attract more short-term investors, which partially offsets the stress.

7.3 Takeaway

This paper gives a clean microfoundation for why external finance becomes especially expensive in downturns. The primary-market pricing wedge is not an incidental detail; it is an economically meaningful friction. For policy, the message is that supporting secondary-market prices alone may not be enough if the real friction sits inside the primary-market issuance process.